

Session_11

Q.1 Declare an integer variable called likes and a pointer variable called ptrLikes; assign likes a value, point ptrLikes to likes, and print both the value and the address stored in ptrLikes.

A.1 #include <stdio.h>

```
int main()
{
    int likes = 500;

    int *ptrLikes;

    ptrLikes = &likes;

    printf("Value of likes = %d\n", likes);

    printf("Address stored in ptrLikes = %p\n", (void *)ptrLikes);

    return 0;
}
```

- likes is an integer variable containing 500.
- ptrLikes is a pointer variable.
- ptrLikes = &likes; stores the address of likes in ptrLikes.
- %p is used to print a memory address.

Q.2 Write a function swapPlaylistCounts(int *a, int *b) that swaps the number of songs in two Spotify playlists using pointers, then call the function in main and print the swapped values.

A.2 #include <stdio.h>

```
void swapPlaylistCounts(int *a, int *b)
{
    int temp;

    temp = *a;
    *a = *b;
    *b = temp;
}

int main() {
    int playlist1 = 50;
```

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```
int playlist2 = 80;

printf("Before swapping:\n");
printf("Playlist 1 = %d songs\n", playlist1);
printf("Playlist 2 = %d songs\n", playlist2);

swapPlaylistCounts(&playlist1, &playlist2);

printf("\nAfter swapping:\n");
printf("Playlist 1 = %d songs\n", playlist1);
printf("Playlist 2 = %d songs\n", playlist2);
return 0;

}
```

Q.3 Create a function `incrementFollowers(int *followers, int n)` that increases each follower count in an array (representing Instagram followers for 5 friends) by 100 using pointer arithmetic, then print the updated counts.

A.3 `#include <stdio.h>`

```
int main() {
    int orders[5] = {250, 180, 320, 150, 400};
    int *ptr;

    ptr = orders;

    for (int i = 0; i < 5; i++)
    {
        printf("Order amount = %d, Address = %p\n",
            *(ptr + i), (void *) (ptr + i));
    }
    return 0;
}
```

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Q.4 Create a function `incrementFollowers(int *followers, int n)` that increases each follower count in an array (representing Instagram followers for 5 friends) by 100 using pointer arithmetic, then print the updated counts.

A.4 `#include <stdio.h>`

```
void incrementFollowers(int *followers, int n)
{
    for (int i = 0; i < n; i++)
    {
        *(followers + i) = *(followers + i) + 100;
    }
}

int main()
{
    int followers[5] = {1200, 2500, 1800, 3200, 1500};

    incrementFollowers(followers, 5);

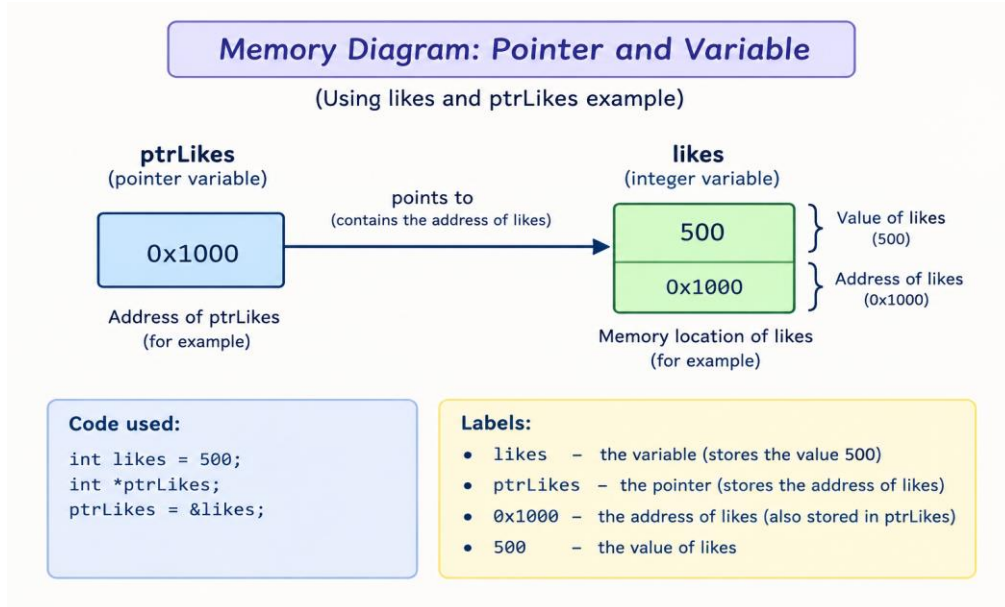
    printf("Updated follower counts:\n");
    for (int i = 0; i < 5; i++)
    {
        printf("Friend %d = %d followers\n", i + 1, followers[i]);
    }

    return 0;
}
```

Q.5 Draw a simple memory diagram (on paper or using any drawing tool) showing how a pointer references the address of a variable, using the variables from your likes and ptrLikes example;

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label the variable, pointer, and address.



A.5